Calculus II Assignment 9

Part I. Regular Problems

1. Compute the partial derivatives.
   (a) f_{xx} and f_{zz} for $(x, y, z) \neq (0, 0, 0)$ if $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$
   (b) f_{xyz} if $f(x, y, z) = xy + yz + zx$
   (c) $g_{xxy} - g_{xyx}$ if g is three times continuously differentiable from $\mathbb{R}^3$ to $\mathbb{R}$
   (d) h_{x_j} and $h_{x_j x_k}$ if $h(\mathbf{X}) = \mathbf{A} \cdot \mathbf{X}$, where $\mathbf{A} = (a_1, \ldots, a_n)$ is constant and $\mathbf{X} = (x_1, \ldots, x_n)$.
2. Let $f(x, y, z) = x^2 + xy + \frac{1}{2}y^2 + 2yz + z^3$.
   (a) Show that ∇f is the zero vector at $\mathbf{0}$ and at one other point $\mathbf{A}$.
   (b) Show that f has a local minimum at $\mathbf{A}$.
   (c) Find points arbitrarily near $(0, 0, 0)$ where f is positive and where f is negative, to show that f has a saddle at $(0, 0, 0)$.
3. Show that the maximum value of the function
   $$Q(x, y) = 3x^2 + 2xy + 3y^2$$
   on the unit circle $x^2 + y^2 = 1$ is 4.
4. Evaluate the integrals if $R = [0, 1] \times [0, 1]$.
   (a) $\iint_R (x^3 + y^2)\, dA$
   (b) $\iint_R ye^{xy} dA$
   (c) $\iint_R (xy)^2 \cos x^3 dA$
   (d) $\iint_R \log[(x + 1)(y + 1)] dA$
5. Evaluate $\iint_R f(x, y) dx dy$ for the indicated functions and rectangles.
   (a) $f(x, y) = (x + 2y)^2; R = [-1, 2] \times [0, 2]$
   (b) $f(x, y) = y^3 \cos^2 x; R = [-\pi/2, \pi] \times [1, 2]$
   (c) $f(x, y) = xy^3 e^{x^2 y^2}; R = [1, 3] \times [1, 2]$
   (d) $f(x, y) = xy + x/(y + 1); R = [1, 4] \times [1, 2]$

Part II. Group Problems

1. Let $f(x,y) = (x-1)^2 + 2(y-2)^2 + (y-2)^3 = 1 - 2x + x^2 + 4y - 4y^2 + y^3$.

   (a) Calculate ∇f and find the two points where $\nabla f = \mathbf{0}$.

   (b) Calculate the Hessian matrix $\mathcal{H}f$ of f at the two points in part (a).

   (c) Determine whether f has a local maximum, minimum, or saddle at the points you found.

2. Evaluate the double integrals:

   (a) $\iint_R f(x,y)dA$, where

      i. $f(x,y) = 7$ and $R = [3,5] \times [-1,9]$

      ii. $f(x,y) = x$ and $R = [0,1] \times [0,1]$

   (b) $\iint_R (x^2y^2 + x)\, dydx$, where R is the rectangle $[0,2] \times [-1,0]$.

   (c) $\iint_R \left(|y| \cos \frac{1}{4}\pi x\right) dydx$, where R is the rectangle $[0,2] \times [-1,0]$.